

LibrarySystem standard_toolkit

$[PERSON, ISBN, TITLE]$

$GENRE ::= fiction \mid nonfiction \mid science \mid history \mid biography$

$REPORT ::= ok$
 $\mid bookNotFound$
 $\mid memberNotFound$
 $\mid bookAlreadyExists$
 $\mid memberAlreadyExists$
 $\mid alreadyBorrowed$
 $\mid notBorrowed$
 $\mid loanLimitReached$

$maxLoans : \mathbb{N}_1$
$maxLoans = 5$

$Library$ $books : \mathbb{P} ISBN$ $members : \mathbb{P} PERSON$ $title : ISBN \rightarrow TITLE$ $genre : ISBN \rightarrow GENRE$ $loans : ISBN \rightarrow PERSON$ $waitlist : ISBN \rightarrow seq PERSON$
$dom\ title = books$ $dom\ genre = books$ $dom\ loans \subseteq books$ $ran\ loans \subseteq members$ $dom\ waitlist \subseteq books$ $\forall m : members \bullet \#\{b : dom\ loans \mid loans(b) = m\} \leq maxLoans$

$borrowedBy : PERSON \times \mathbb{P}(ISBN \times PERSON) \rightarrow \mathbb{P} ISBN$
$\forall m : PERSON; ln : \mathbb{P}(ISBN \times PERSON) \bullet$ $borrowedBy(m, ln) = \{b : dom\ ln \mid (b, m) \in ln\}$

$availableBooks : \mathbb{P} ISBN \times \mathbb{P}(ISBN \times PERSON) \rightarrow \mathbb{P} ISBN$
$\forall bs : \mathbb{P} ISBN; ln : \mathbb{P}(ISBN \times PERSON) \bullet$ $availableBooks(bs, ln) = bs \setminus dom\ ln$

$loanCount : PERSON \times \mathbb{P}(ISBN \times PERSON) \rightarrow \mathbb{N}$
$\forall m : PERSON; ln : \mathbb{P}(ISBN \times PERSON) \bullet$ $loanCount(m, ln) = \#\{b : \text{dom } ln \mid (b, m) \in ln\}$
$booksByGenre : GENRE \times \mathbb{P} ISBN \times (ISBN \rightarrow GENRE) \rightarrow \mathbb{P} ISBN$
$\forall g : GENRE; bs : \mathbb{P} ISBN; gn : ISBN \rightarrow GENRE \bullet$ $booksByGenre(g, bs, gn) = \{b : bs \mid b \in \text{dom } gn \wedge gn(b) = g\}$

$InitLibrary$
$Library'$
$books' = \emptyset$ $members' = \emptyset$ $title' = \emptyset$ $genre' = \emptyset$ $loans' = \emptyset$ $waitlist' = \emptyset$

$AddBook$
$\Delta Library$ $isbn? : ISBN$ $t? : TITLE$ $g? : GENRE$
$isbn? \notin books$ $books' = books \cup \{isbn?\}$ $title' = title \cup \{isbn? \mapsto t?\}$ $genre' = genre \cup \{isbn? \mapsto g?\}$ $waitlist' = waitlist \cup \{isbn? \mapsto \langle \rangle\}$ $loans' = loans$ $members' = members$

$RemoveBook$
$\Delta Library$ $isbn? : ISBN$
$isbn? \in books$ $isbn? \notin \text{dom } loans$ $books' = books \setminus \{isbn?\}$ $title' = \{isbn?\} \triangleleft title$ $genre' = \{isbn?\} \triangleleft genre$ $waitlist' = \{isbn?\} \triangleleft waitlist$ $loans' = loans$ $members' = members$

<i>RegisterMember</i>
$\Delta\text{Library}$
$p? : \text{PERSON}$
$p? \notin \text{members}$ $\text{members}' = \text{members} \cup \{p?\}$ $\text{books}' = \text{books}$ $\text{title}' = \text{title}$ $\text{genre}' = \text{genre}$ $\text{loans}' = \text{loans}$ $\text{waitlist}' = \text{waitlist}$

<i>RemoveMember</i>
$\Delta\text{Library}$
$p? : \text{PERSON}$
$p? \in \text{members}$ $p? \notin \text{ran loans}$ $\text{members}' = \text{members} \setminus \{p?\}$ $\text{books}' = \text{books}$ $\text{title}' = \text{title}$ $\text{genre}' = \text{genre}$ $\text{loans}' = \text{loans}$ $\text{waitlist}' = \text{waitlist}$

<i>BorrowBook</i>
$\Delta\text{Library}$
$\text{isbn?} : \text{ISBN}$
$p? : \text{PERSON}$
$\text{isbn?} \in \text{books}$ $p? \in \text{members}$ $\text{isbn?} \notin \text{dom loans}$ $\text{loanCount}(p?, \text{loans}) < \text{maxLoans}$ $\text{loans}' = \text{loans} \cup \{\text{isbn?} \mapsto p?\}$ $\text{books}' = \text{books}$ $\text{members}' = \text{members}$ $\text{title}' = \text{title}$ $\text{genre}' = \text{genre}$ $\text{waitlist}' = \text{waitlist}$

<i>ReturnBook</i>
$\Delta Library$
$isbn? : ISBN$
$p? : PERSON$
$isbn? \in \text{dom } loans$
$loans(isbn?) = p?$
$loans' = \{isbn?\} \triangleleft loans$
$books' = books$
$members' = members$
$title' = title$
$genre' = genre$
$waitlist' = waitlist$

<i>SearchByGenre</i>
$\Xi Library$
$g? : GENRE$
$result! : \mathbb{P} ISBN$
$books \neq \emptyset$
$result! = \text{booksByGenre}(g?, books, genre)$

<i>GetMemberBooks</i>
$\Xi Library$
$p? : PERSON$
$result! : \mathbb{P} ISBN$
$p? \in members$
$result! = \text{borrowedBy}(p?, loans)$

<i>Success</i>
$r! : REPORT$
$r! = ok$

<i>BookNotFoundError</i>
$\Xi Library$
$r! : REPORT$
$r! = \text{bookNotFound}$

<i>MemberNotFoundError</i>
$\Xi Library$
$r! : REPORT$
$r! = \text{memberNotFound}$

<i>BookAlreadyExistsError</i>
$\exists \text{Library}$ $r! : \text{REPORT}$ $\text{isbn?} : \text{ISBN}$
$\text{isbn?} \in \text{books}$ $r! = \text{bookAlreadyExists}$

<i>MemberAlreadyExistsError</i>
$\exists \text{Library}$ $r! : \text{REPORT}$ $p? : \text{PERSON}$
$p? \in \text{members}$ $r! = \text{memberAlreadyExists}$

<i>AlreadyBorrowedError</i>
$\exists \text{Library}$ $r! : \text{REPORT}$ $\text{isbn?} : \text{ISBN}$
$\text{isbn?} \in \text{dom loans}$ $r! = \text{alreadyBorrowed}$

<i>NotBorrowedError</i>
$\exists \text{Library}$ $r! : \text{REPORT}$ $\text{isbn?} : \text{ISBN}$
$\text{isbn?} \notin \text{dom loans}$ $r! = \text{notBorrowed}$

<i>LoanLimitError</i>
$\exists \text{Library}$ $r! : \text{REPORT}$ $p? : \text{PERSON}$
$\text{loanCount}(p?, \text{loans}) \geq \text{maxLoans}$ $r! = \text{loanLimitReached}$

$\text{AddBookOk} == \text{AddBook} \wedge \text{Success}$

$\text{TotalAddBook} == \text{AddBookOk} \vee \text{BookAlreadyExistsError}$

<i>BookOnLoan</i>
<i>Library</i> <i>isbn?</i> : <i>ISBN</i>
<i>isbn?</i> ∈ dom <i>loans</i>

$BookNotOnLoan == \neg BookOnLoan$

$BorrowAndQuery == BorrowBook \wp GetMemberBooks$

$BorrowBookOk == BorrowBook \wedge Success$

$TotalBorrowBook == BorrowBookOk \vee BookNotFound\!Error$
 $\vee AlreadyBorrowed\!Error \vee LoanLimit\!Error$

$ReturnBookOk == ReturnBook \wedge Success$

$TotalReturnBook == ReturnBookOk \vee NotBorrowed\!Error$

<i>ConcreteLibrary</i>
<i>cbooks</i> : seq <i>ISBN</i> <i>cmembers</i> : seq <i>PERSON</i> <i>ctitle</i> : seq(<i>ISBN</i> × <i>TITLE</i>) <i>cgenre</i> : seq(<i>ISBN</i> × <i>GENRE</i>) <i>cloans</i> : seq(<i>ISBN</i> × <i>PERSON</i>) <i>cwaitlist</i> : seq(<i>ISBN</i> × seq <i>PERSON</i>)
$\forall i, j : \text{dom } cbooks \bullet$ $i \neq j \Rightarrow cbooks(i) \neq cbooks(j)$ $\forall i, j : \text{dom } cmembers \bullet$ $i \neq j \Rightarrow cmembers(i) \neq cmembers(j)$ $\forall i : \text{dom } cloans \bullet$ $first(cloans(i)) \in \text{ran } cbooks$ $\forall i : \text{dom } cloans \bullet$ $second(cloans(i)) \in \text{ran } cmembers$ $\forall m : \text{ran } cmembers \bullet$ $\#\{i : \text{dom } cloans \mid second(cloans(i)) = m\} \leq maxLoans$

<i>Retrieve</i>
<i>Library</i>
<i>ConcreteLibrary</i>
<i>books</i> = ran <i>cbooks</i> <i>members</i> = ran <i>cmembers</i> <i>title</i> = ran <i>ctitle</i> <i>genre</i> = ran <i>cgenre</i> <i>loans</i> = ran <i>cloans</i> <i>waitlist</i> = ran <i>cwaitlist</i>

<i>ConcreteInitLibrary</i>
<i>ConcreteLibrary</i> '
<i>cbooks</i> ' = $\langle \rangle$ <i>cmembers</i> ' = $\langle \rangle$ <i>ctitle</i> ' = $\langle \rangle$ <i>cgenre</i> ' = $\langle \rangle$ <i>cloans</i> ' = $\langle \rangle$ <i>cwaitlist</i> ' = $\langle \rangle$

<i>ConcreteBorrowBook</i>
Δ <i>ConcreteLibrary</i>
<i>isbn?</i> : <i>ISBN</i> <i>p?</i> : <i>PERSON</i>
<i>isbn?</i> $\in$ ran <i>cbooks</i> <i>p?</i> $\in$ ran <i>cmembers</i> $\neg (\exists i : \text{dom } \textit{cloans} \bullet \textit{first}(\textit{cloans}(i)) = \textit{isbn?})$ $\#\{i : \text{dom } \textit{cloans} \mid \textit{second}(\textit{cloans}(i)) = \textit{p?}\} < \textit{maxLoans}$ <i>cloans</i> ' = <i>cloans</i> $\cap \langle (\textit{isbn?}, \textit{p?}) \rangle$ <i>cbooks</i> ' = <i>cbooks</i> <i>cmembers</i> ' = <i>cmembers</i> <i>ctitle</i> ' = <i>ctitle</i> <i>cgenre</i> ' = <i>cgenre</i> <i>cwaitlist</i> ' = <i>cwaitlist</i>

ConcreteReturnBook ΔConcreteLibrary $isbn? : ISBN$ $p? : PERSON$ $\exists i : \text{dom } cloans \bullet cloans(i) = (isbn?, p?)$ $\text{ran } cloans' = \text{ran } cloans \setminus \{(isbn?, p?)\}$ $\forall i : \text{dom } cloans' \bullet cloans'(i) \in \text{ran } cloans \setminus \{(isbn?, p?)\}$ $cbooks' = cbooks$ $cmembers' = cmembers$ $ctitle' = ctitle$ $cgenre' = cgenre$ $cwaitlist' = cwaitlist$
ConcreteSearchByGenre ΞConcreteLibrary $g? : GENRE$ $result! : \mathbb{P} ISBN$ $\text{ran } cbooks \neq \emptyset$ $result! = \{i : \text{dom } cgenre \mid \text{second}(cgenre(i)) = g? \bullet \text{first}(cgenre(i))\}$